

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“Jnana Sangama”, Belagavi-590018



Submitted in Partial fulfillment of requirements for the Mobile
Application Development Laboratory (18CSMP68)

Mini Project Report On

“SMART AND HASSLE FREE CHECKOUT IN SUPERMARKET ”

*Submitted in the partial fulfillment for the award of Bachelor of Engineering
degree in COMPUTER SCIENCE & ENGINEERING*

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2022-2023

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CERTIFICATE

This is to certify that the Project Work entitled “**SMART AND HASSLE FREE CHECKOUT IN SUPERMARKET**” is the bonafide work carried out by **ABHISHEK M (4AD20CS002)** and **DHANUSH GOWDA G P (4AD20CS023)** in partial fulfillment for the award of degree of Bachelor of Engineering in Computer Science and Engineering from Visvesvaraya Technological University, Belagavi during the year 2022-2023.

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ABSTRACT

The checkout process in supermarkets is an essential yet time-consuming task that often leads to customer dissatisfaction and operational inefficiencies. To address these challenges, this project introduces a smart and hassle-free checkout system designed specifically for supermarkets, utilizing the Android platform.

The proposed system leverages cutting-edge technologies such as barcode scanning, mobile payments, and cloud-based data management to streamline the checkout experience. Customers can use their smartphones equipped with the dedicated Android application to scan product barcodes, view real-time prices, and add items to their virtual shopping carts. The application incorporates advanced algorithms to ensure accurate barcode scanning, enhancing the speed and efficiency of the checkout process.

Additionally, the Android application integrates secure mobile payment options, allowing customers to conveniently complete transactions without the need for traditional cash or credit cards. Payment information is encrypted and transmitted securely, guaranteeing the safety of sensitive data. The integration of popular mobile payment platforms further enhances user convenience and flexibility. To facilitate seamless synchronization and data management, the project employs cloud-based technology.

The system maintains a centralized database hosted on the cloud, which stores information regarding products, prices, and customer transactions. This enables real-time updates, ensuring accurate stock management, pricing, and inventory control. The cloud-based approach also allows for easy scalability and adaptability, accommodating the changing demands of a dynamic supermarket environment.

CONTENTS

SL NO	CHAPTERS	PAGE NO
1	CHAPTER 1: INTRODUCTION	1-7
	1.1 Introduction	1
	1.2 Android	2
	1.3 Android Architecture	3-4
	1.4 Android Studio	5
	1.5 Objective	5-6
	1.6 Methodology	6
	1.7 About The Project	6
	1.8 Problem Statement	7
	1.9 Aim And Objectives	7
	1.10 Scope	7
2	CHAPTER 2: SYSTEM REQUIREMENTS	8-12
	2.1 Functional Requirements	8-10
	2.2 System Requirements	10-11
	2.3 Hardware Requirements	11
	2.4 Software Requirements	12
3	CHAPTER 3: SYSTEM DESIGN	13-19
4	CHAPTER 4: SYSTEM IMPLEMENTATION	20-21
	4.1 Project Description	20
	4.2 Type of Implementation	21
5	CHAPTER 5: SNAPSHOTS	22-24
6	CONCLUSION	25
7	REFERENCES	26

Chapter 1

INTRODUCTION

1.1 Introduction:

Smart and hassle-free checkout in supermarket is a mobile application. The customer can scan the product and pay using a QR code. This makes the work simple for customers so that they need not wait for long hours in a queue. One advantage of this mobile application is that there is no need for lengthy queue and in this busy schedule of everyone this mobile application makes the work easy and it can save time.

Two-dimensional matrix codes also known as the QR Code (Quick Response Code) , on other side, is advancement over the Barcode Systems and is used for various commercial purposes. The QR Code stores information in two dimensions that can be scanned and read by an application known as the QRCode Reader. The QR code can be used to store various form of information like images, videos, price and all. The objective of this project is to propose a real time capturing system for consumer supplies using Quick Response (QR) code in an Android smart phone. Using multiplexing and demultiplexing process encode and decode the information from single QR code with special symbols and split the data back to their QR Code pattern where these QR Code pattern can be read by Android smart phones. The customer will have to put the product in front of QR Code scanner; it scans the Code and saves the information of the product in the cart

1.2Android:

Android is a mobile operating system (OS) currently developed by Google, based on the Linux kernel and designed primarily for touchscreen mobile devices such as smartphones and tablets. Android's user interface is mainly based on direct manipulation, using touch gestures that loosely correspond to real-world actions, such as swiping, tapping and pinching, to manipulate on-screen objects, along with a virtual keyboard for text input. In addition to touchscreen devices, Google has further developed Android TV for televisions, Android Auto for cars, and Android Wear for wrist watches, each with a specialized user interface. Variants of Android are also used on notebooks, game consoles, digital cameras, and other electronics. Initially developed by Android, Inc., which Google bought in 2005, Android was unveiled in 2007, along with the founding of the Open Handset Alliance – a consortium of hardware, software, and telecommunication companies devoted to advancing open standards for mobile devices. As of July 2013, the Google Play store has had over one million Android applications ("apps") published, and over 50 billion applications downloaded. An April–May 2013 survey of mobile application developers found that 71% of developers create applications for Android, and a 2015 survey found that 40% of full-time professional developers see Android as their priority target platform.



1.3 Android Architecture:

We studied the Android system architecture. Android system is a Linux-based system, Use of the software stack architecture design patterns. As shown in Figure, the Android architecture consists of four layers: Linux Kernel, Libraries and Android runtime, Application framework and Applications. Each layer of the lower encapsulation, while providing call interface to the upper.

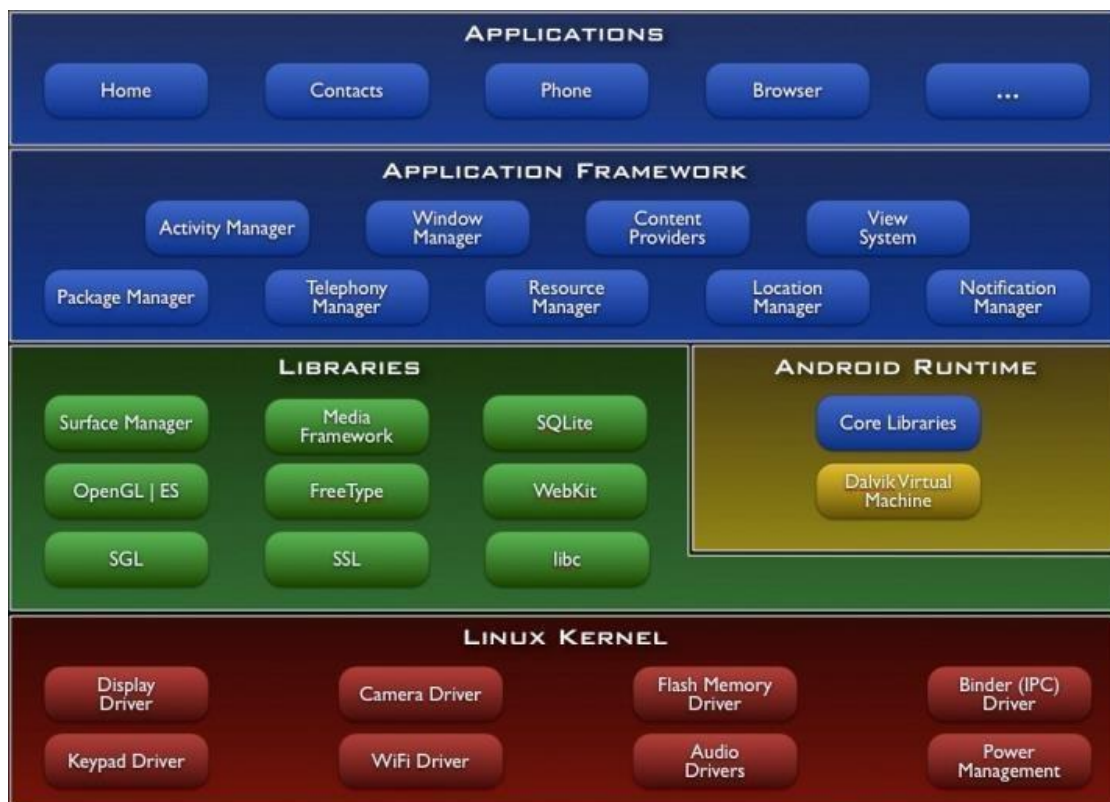


Figure 1.3 :Android Architecture

Libraries:

On top of Linux kernel there is a set of libraries including open -source Web browser engine Web Kit, well known library libc, SQLite database which is a useful repository for storage and sharing of application data, libraries to play and record audio and video, SSL libraries responsible for Internet security etc.

Android Runtime:

This is the third section of the architecture and available on the second layer from the bottom. This section provides a key component called Dalvik Virtual Machine which is a kind of Java Virtual Machine specially designed and optimized for Android.

The Dalvik VM makes use of Linux core features like memory management and multithreading, which is intrinsic in the Java language. The Dalvik VM enables every Android

The Android runtime also provides a set of core libraries which enable Android application developers to write Android applications using standard Java programming language.

Application Framework:

The Application Framework layer provides many higher-level services to applications in the form of Java classes. Application developers are allowed to make use of these services in their applications, as well as set up.

Applications:

You will find all the Android application at the top layer. You will write your application to be installed on this layer only. Examples of such applications are Contacts Books, Browser, and Games etc.

Android UI:

An Android application user interface is everything that the user can see and interact with. Installation steps of the developing environment

- Step 1: install the Java virtual machine JDK version -7
- Step 2: install the Android SDK: first download the Android SDK
- Download address: <http://developer.android.com/sdk/index.html>

- Input SDK tools path in the SDK location: D: \ android \ software \ android SDK– Windows and click OK
- The Android environment is set up successfully.

1.4 Android Studio:

Android Studio is an integrated development environment (IDE) for developing for the Android platform. It was announced on May 16, 2013 at the Google I/O conference. Android Studio is freely available under the Apache License 2.0. Android Studio was in early access preview stage starting from version 0.1 in May 2013, then entered beta stage starting from version 0.8 which was released in June 2014. The first stable build was released in December 2014, starting from version 1.0. Based on JetBrains' IntelliJ IDEA software, Android Studio is designed specifically for Android development. It is available for download on Windows, Mac OS X and Linux, and replaced Eclipse Android Development Tools (ADT) as Google's primary IDE for native Android application development.



Figure 1.3: Android studio logo

1.5 Objective:

There are 5 objectives in this project:

- To setup Android software development kit
- To write a program that can do various tasks such as running media(audio&video), displaying information, sending notifications, Bitmapping, asynchronous multi-tasking.

- Crash handling and debugging.
- Generating signed APK.

Upload application on Google Playstore.

1.6 Methodology:

This project is made by using Android studio, Virtual emulator and Photoshop. The programming languages used for building the application are Java, XML and C++. User interface is handled using XML codes. Backend programming is handled mainly through set of java codes. Native libraries are accessed through set of predefined C++ codes.

The other libraries required are:

- Android SDK tools
- Android SDK platform tools
- Android SDK build tools
- SDK platform
- Google APIs
- Google APIs ARM EABI v7a System Image

Intel x86 Emulator Accelerator (HAXM installer)

1.7 About The Project:

The inspiration for doing the project was essentially a enthusiasm for undertaking a difficult venture in a fascinating territory of research. This project has given me the opportunity to learn and develop a android app from scratch. This app can be used by admin and the customer, floor manager to that add products available in that particular floor.

1.8 Problem statement:-

Smart and hassle-free checkout in supermarket is a mobile application. The customer can scan the product and pay using a QR code. This makes the work simple for customers so that they need not wait for long hours in a queue. One advantage of this mobile application is that there is no need for lengthy queue and in this busy schedule of everyone this mobile application makes the work easy and it can save time .

1.9 Aim and Objectives

Aim:

- To significantly save time, effort, and money.
- To modify the learning sessions in accordance with the needs
- To simplify the process of review and feedback.

Objectives:

The objective of the Android System is to provide better information for the users of this system better results for their maintenance in customer and product grading details.

- The inspiration for doing this project was essentially an enthusiasm for undertaking a difficult venture in a fascinating territory of research.
- This project has given me the opportunity to learn and develop a android app from scratch.
- This app can be used by admin and the customer, floor manager to add products available in that floor..

1.10 Scope

- This app can be used by employees of super markets to add products and generate QR code.
- Also any user interested in buying products from super markets can use this app.

Chapter 2

SYSTEM REQUIREMENTS

The essential information that forms the basis of the product advancement measure is known as the Software Requirement Specification (SRS). SRS records the necessities of a system just as has a portrayal of its critical segments. The concentration in this stage is one of the clients of the framework and not the framework arrangements. The consequence of the necessity particular archive expresses the expectation of the product, properties, and imperatives of the ideal framework. SRS comprises the comprehension among clients and creators concerning the substance of the item that will be made. SRS ought to be exact and thoroughly imply the structure requirements as it makes a monster obligation to the overall advancement plan.

Operating System	Windows 7 or higher
Languages Used	Java, XML
Development Environment (IDE)	Android Studio
Database	SQLite

2.1 Functional Requirements

It defines the functionality of the software and the constraints on it. Functional requirements describe the features, functioning, and usage of a product/system/software from the perspective of the product and its user. The document will not only define the product functions, user characteristics and constraints but also

serve as a basis for software design document.

Modules:

The system comprises of 3 major modules:

- 1.Admin
- 2.Floor Manager
- 3.Customer

Admin: This module contains all the information about the admin. Once the admin is logged in to the app, he or she can approve the registered customer, and even delete the registered customer.

- **Login:** Allow the admin to login to the app with valid credentials.
- **Approve customer/Floor Manager:** Allow the admin to approve the customer and verify their details.
- **Delete customer/Floor Manager:** Allow the admin to delete the customer and floor manager.
- **View customers :** Allow the admin to see the list of the approved customers.
- **View billing:** Allow the admin to view the bill of the customer.
- **Logout:** Allow the admin to logout from the app.

Floor manager: This module consists all the information about the floor manager. Once the floor manager is logged in to the app, he or she can store the products available in the particular floor, He or she can even view the bills of the customer and verify the products.

- **Register:** Allow the floor manager to register with the app with all his details like Name, Username, password, mobile number etc.
- **Login:** Allow the floor manager to login to the app with valid credentials once he is approved by the admin.
- **Add product details:** Allow floor manager to add the information about products and its cost.
- **View feedback:** Allow floor manager to view feedback given by the customers.
- **Logout:** Allow floor manager to logout of the app.

Customer: This module contains all the information about the customers. Once the customers are logged in, he or she can scan, pay and use the app.

- **Register:** Allow the customer to register with the app with all his details like Name, Username, Password, Mail, Mobile number.
- **Login:** Allow the customer to login to the app with valid credentials. Once he is approved by the admin.
- **Scan the products:** Allow the customer to scan the product.
- **Pay:** Allow the customer to pay after he or she scan the product.
- **Give Feedback:** Allow the customer to give feedback.
- **Logout:** Allow the customer to logout of the app.

2.2 Non-functional Requirements

Application Security

The system should be protected in such a manner that one registered user should not be able to access another registered user's information ensuring privacy of information.

Database Security

Direct access to the database by system users should not be allowed so they may examine or query its contents. The application interface should be the only way to access the database.

Maintainability

The application should be designed to make it simple to add new goods and simplify the modification of product information.

Consistency

To minimize the learning curve, the appearance and delivery of the content should be uniform. Every page should have uniform layouts, button placement, and positioning of important elements.

Usability

How challenging it will be to understand and use the system.

Scalability

The number of supported users will primarily depend on the server load, processing power, and memory. The most users possible should be supported.

Availability

It should be accessible 24 hours a day, seven days a week, so that students can utilize it whenever it suits them.

2.3 SYSTEM REQUIREMENTS**Performance Requirements:**

- The app must be interactive and the delays involved must be less. So, in every actionresponse of the app, there are no immediate delays.
- Resource consumption of this application should not reach an amount that renders the mobile device unusable.
- The application should be capable of operating in the background should the user wish to utilize other applications.

2.4 Hardware Requirements

RAM: 4 GB recommended

Processor: i5

Hard disk: 10GB

Hard disk	Minimum of 40GB
RAM	4 GB or higher
Device	Android mobile

2.5 Software Requirements

Operation system: Windows 7/8/10

Front-end design: XML

Back-end design: JAVA

Data Base : SQLite

CHAPTER 3

SYSTEM DESIGN

System architecture:

A system architecture is the conceptual model that defines the structure, behaviour, and more views of a system. An architecture description is a formal description and representation of a system, organized in a way that supports reasoning about the structures and behaviours of the system.

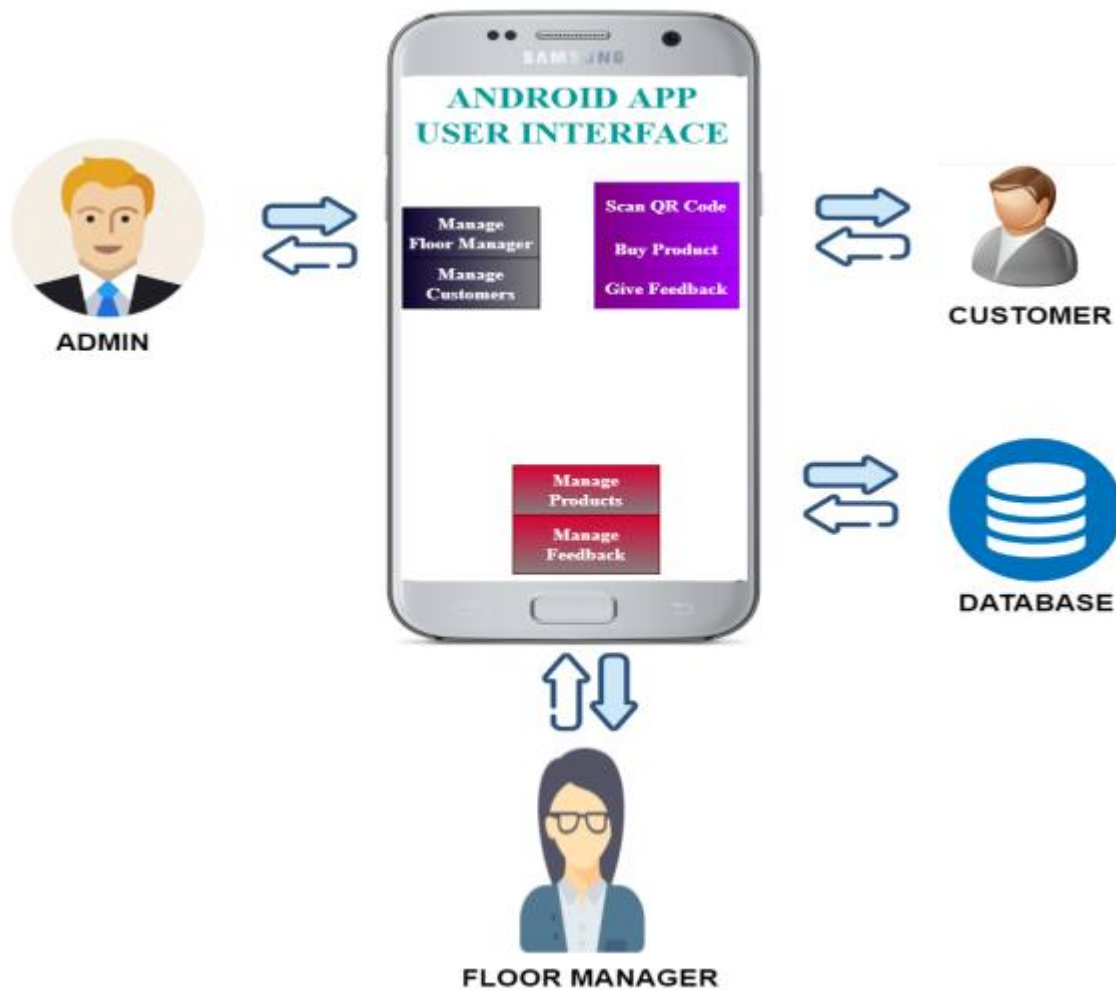


Figure 3.1(a) :System Diagram

3.1 DATA FLOW DIAGRAM

A data flow diagram (DFD) maps out the flow of information for any process or system. It uses defined symbols like rectangles, circles and arrows, plus short text labels, to show data inputs, outputs, storage points and the routes between each destination. Data flowcharts can range from simple, even hand-drawn process overviews, to in-depth, multi-level DFDs that dig progressively deeper into how the data is handled. They can be used to analyze an existing system or model a new one. Like all the best diagrams and charts, a DFD can often visually “say” things that would be hard to explain in words, and they work for both technical and nontechnical audiences, from developer to CEO.

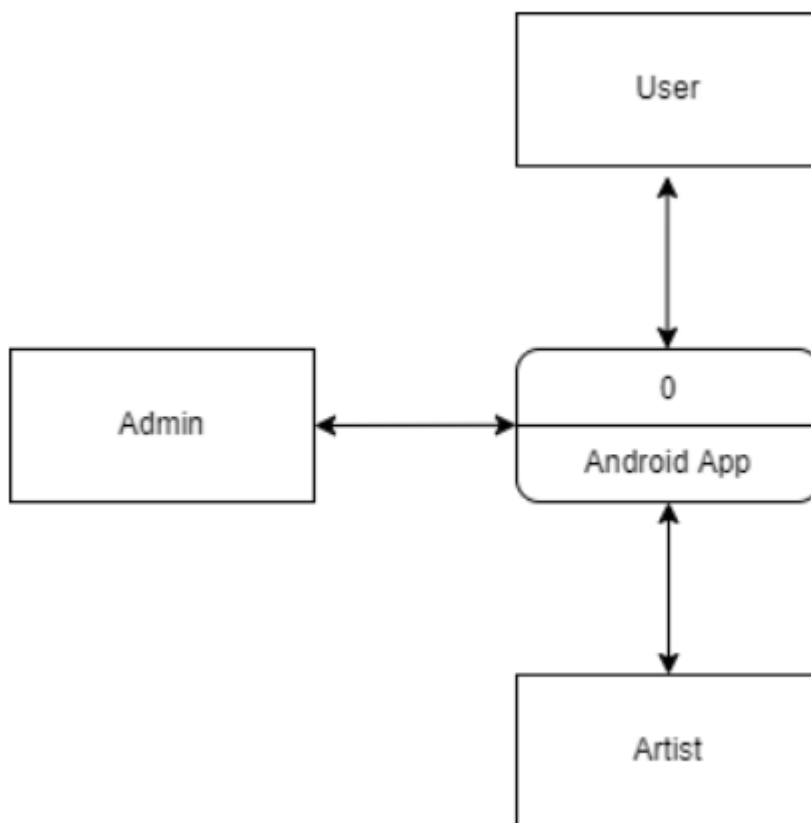


Figure 3.1(b): Data Flow Diagram

Controller Data Flow Diagram (Admin)

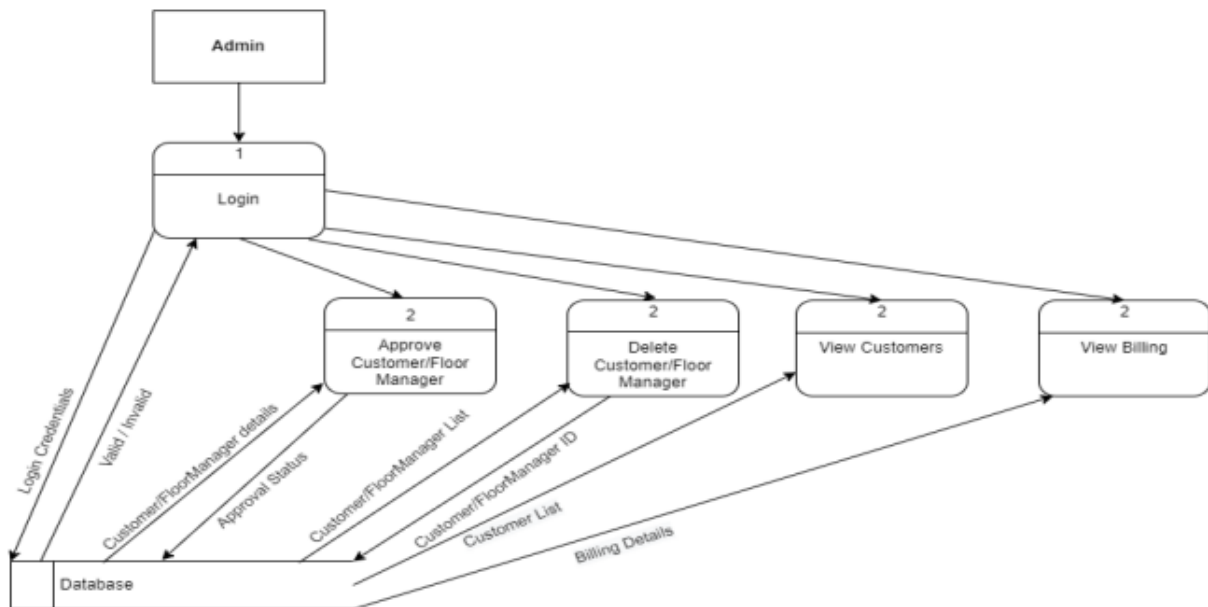


Figure 3.2 Controller Data Flow Diagram

Customer Data Flow Diagram

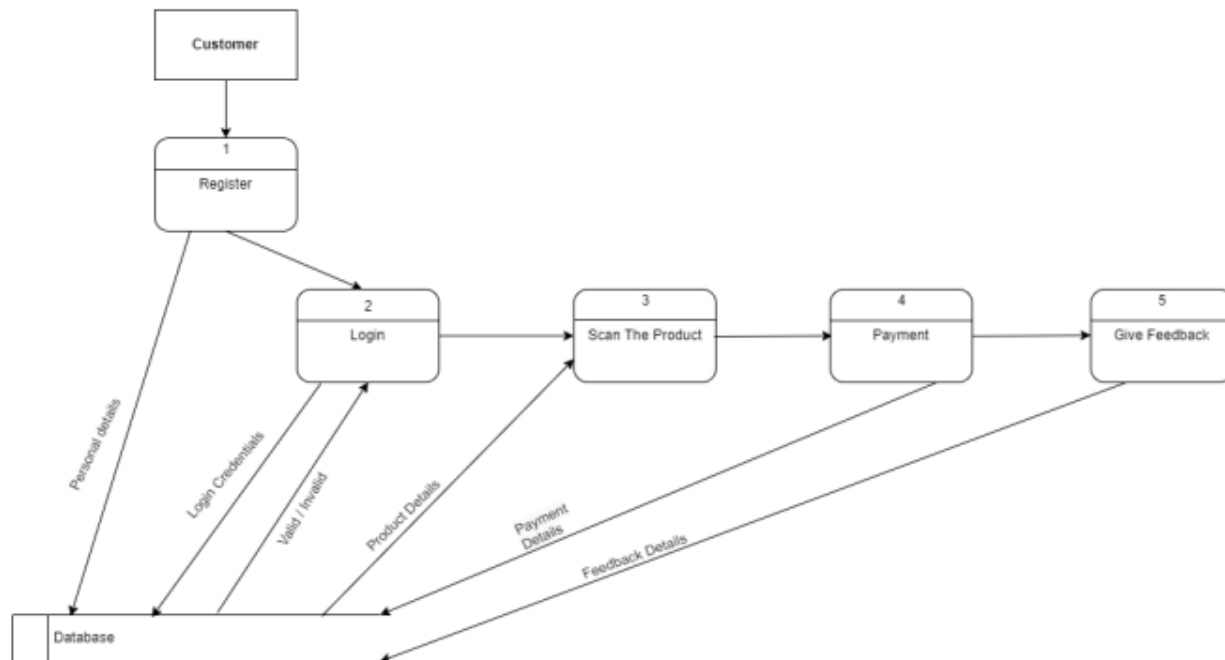


Figure 3.3:Customer Data Flow Diagram

USE CASE DIAGRAMS

A use case diagram is a graphical representation of the user's interaction with the system. It can portray the different types of users of a system and the various ways they interact with the system. Use cases are diagrammed to be easily understood, no matter who is looking at the diagram.

A use case diagram can summarize the details of your system's users (also known as actors) and their interactions with the system.

An effective use case diagram helps to represent:

1. Scenarios in which your system or application interacts with people, organizations, or external systems.
2. Goals that it helps those entities (known as actors) achieve.
3. The scope of your system.

Use Case Diagram for Admin

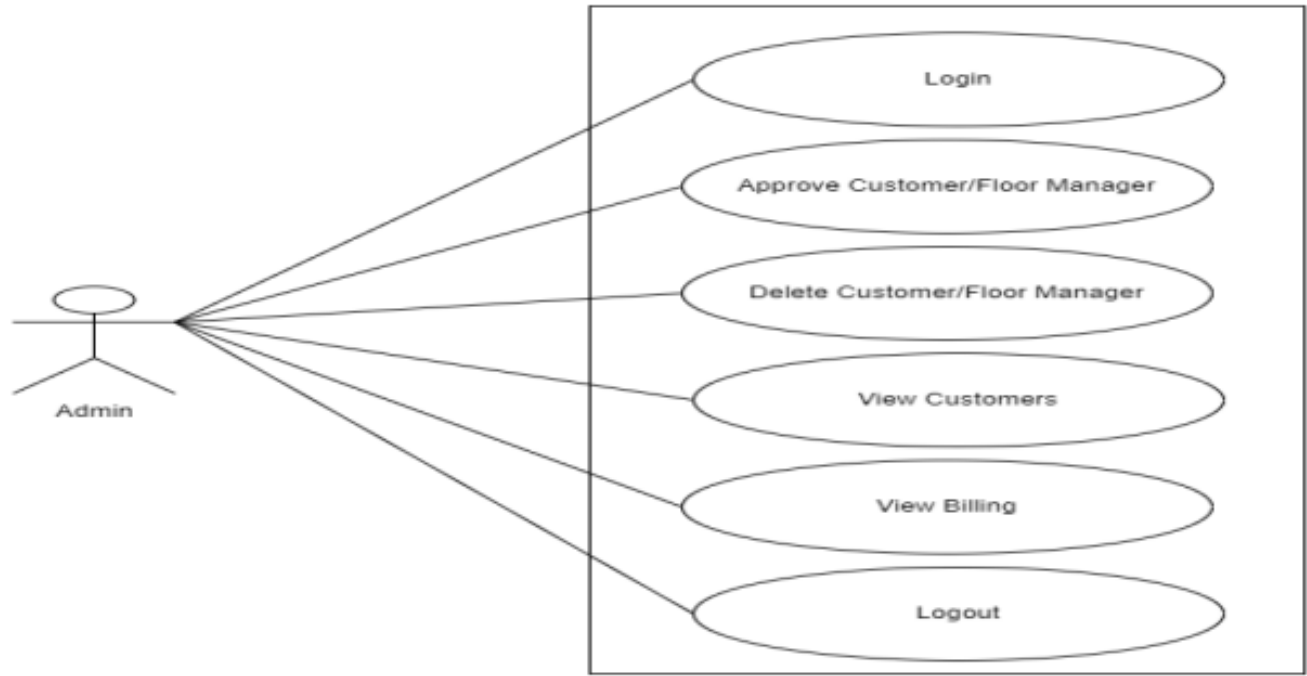


Figure 3.4: use case Diagram for Admin

Use Case Diagram for Floar Manager

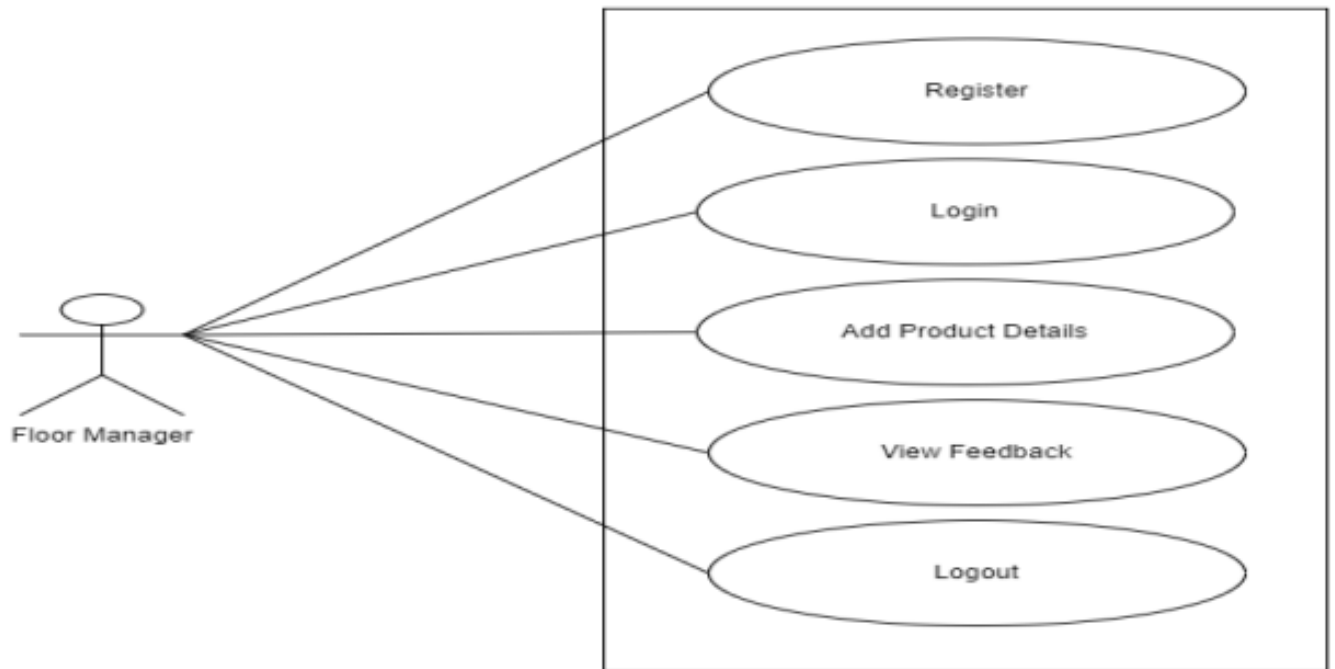


Figure 3.5: use case Diagram for Floar Manager

Use Case Diagram for Customer

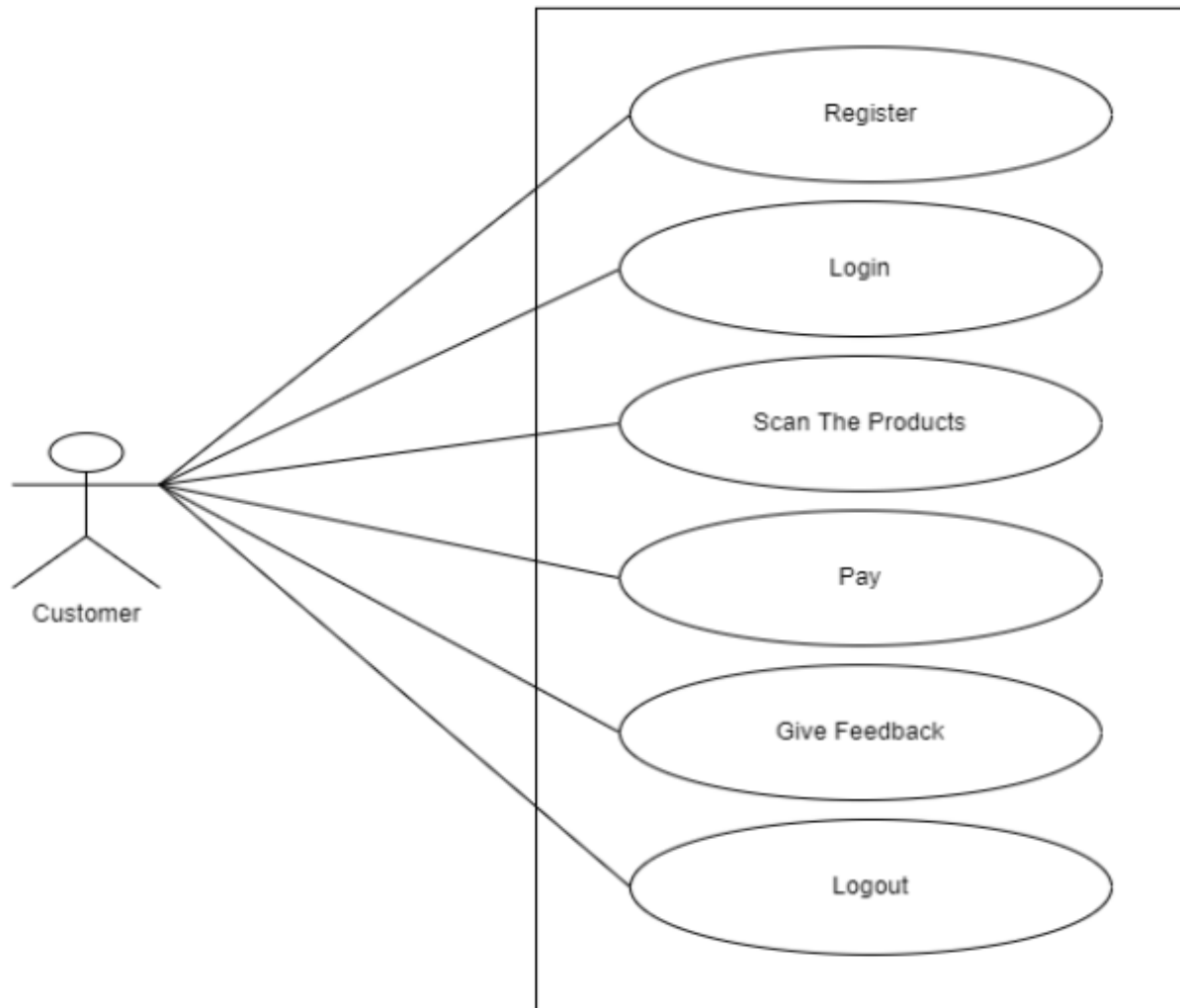


Figure 3.7: use case Diagram for Customer

Chapter 4

SYSTEM IMPLEMENTATION

4.1 Introduction

The goals of the implementation phase is to translate the design of the system produce during the phase , Into coded form in a given programming language, which can then be executed by a computer performing the computation specified by the design the coding phase affects both testing and maintenance profoundly. Well written code can reduce the testing and maintains cost.

A crucial phase in the system lifecycle is the successful implementation of the system design. Implementation simply means converting the system designs into operation. Implementation is the process of bringing the developed system into operational use and providing it to the user.

This stage is considered to be most crucial stage in the development of a successful system since a new system is developed and the users are get information in effective manner Implementation is a stage in which the design is converted into working system that is it is the stage of the project where theoretical design is turned into a working system .The implementation involves careful planning, investing of the current system and its constraint on implementation, design of methods to achieve the changeover.

The Project is implemented in different phases as follows

- First phase includes table design for database module.
- Second phase includes coding for modules.
- Third phase includes the integration of modules.
- Fourth phase includes connection establishment between the front end and back end.
- Fifth phase includes error handling and message generator.

The coding was done with the following characteristics in mind

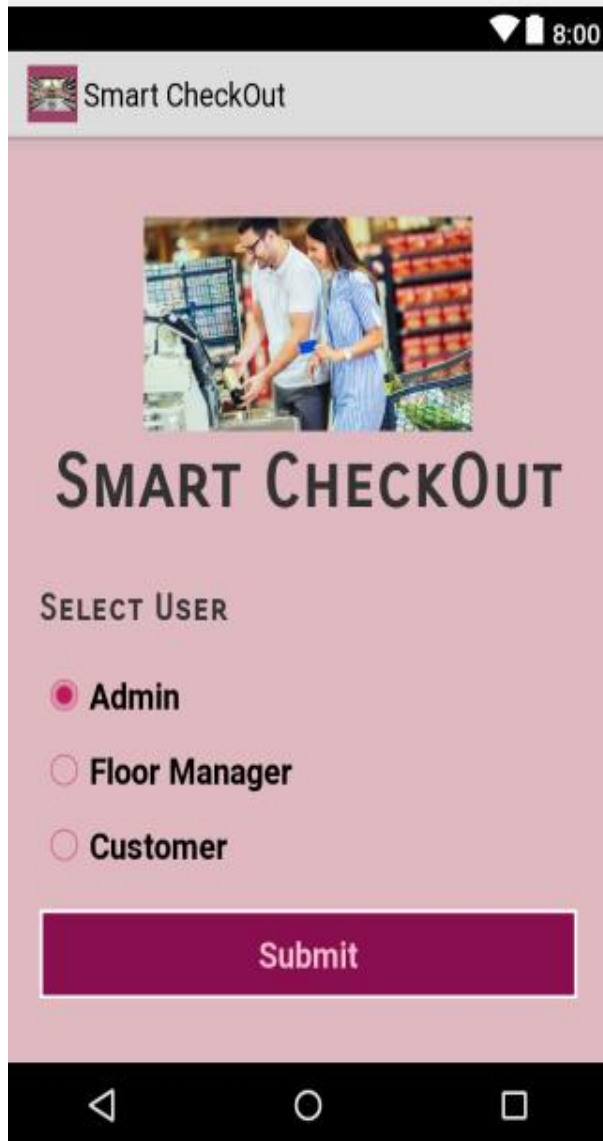
- Code efficiency
- Memory efficiency
- Response time
- Security
- Maintainability
- Efficient and consistent logic.

4.2 PARALLEL CONVERSION TYPE OF IMPLEMENTATION

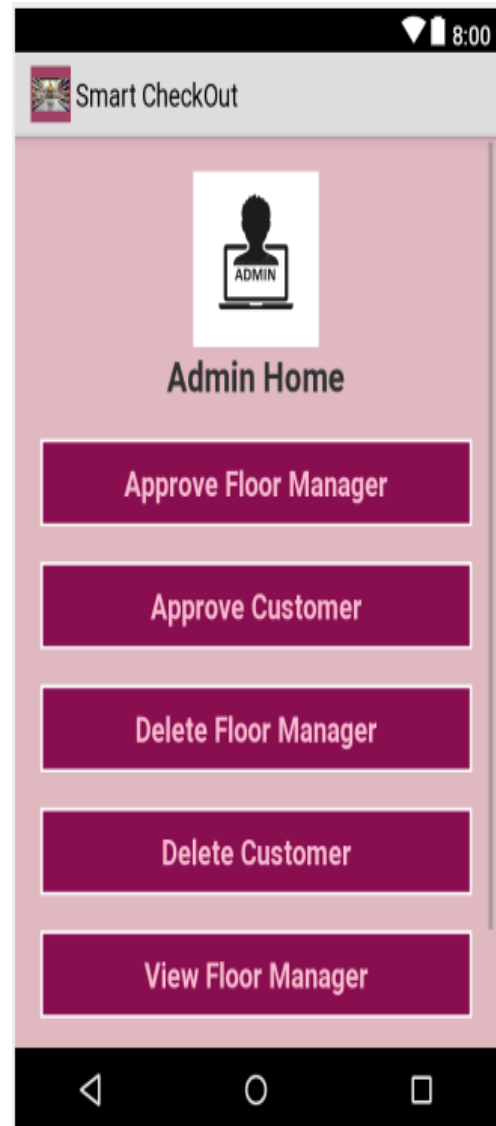
In this type of implementation, both the current system and the proposed system run in parallel. This happens until the user gets complete confidence in the proposed system and hence cuts off the current system.

Chapter 5

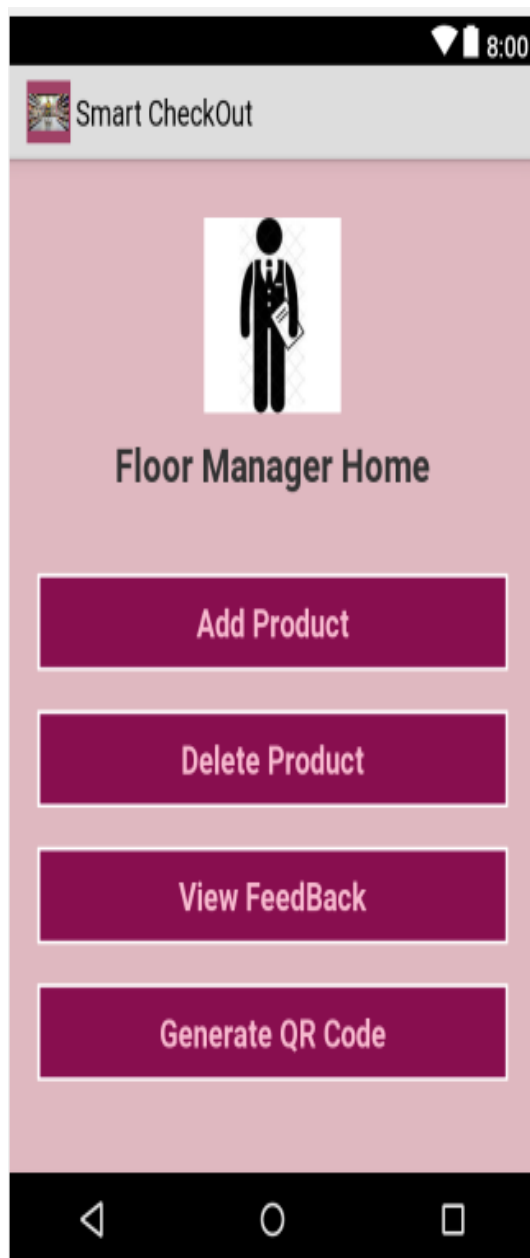
SNAPSHOTS



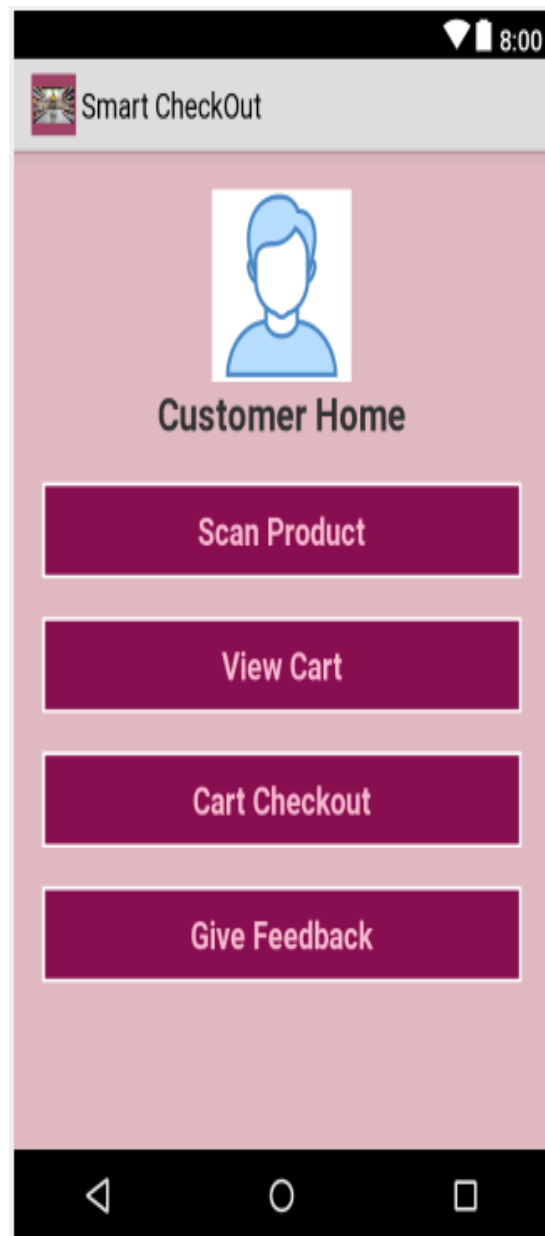
Snapshots 5.1:Home Page



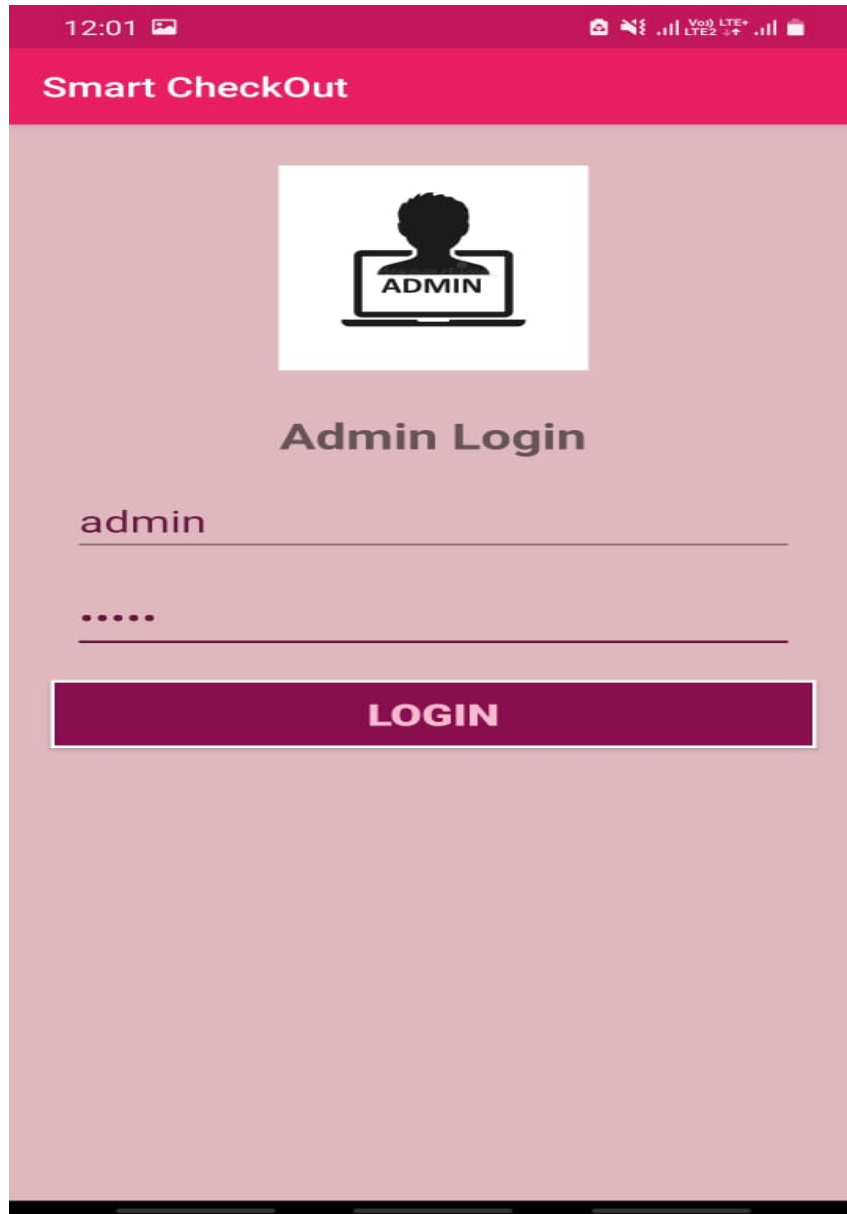
Snapshots 5.2:Admin Page



Snapshots 5.3:Floar Manager Page



Snapshots 5.4:Customer Page



The image shows a mobile application interface for 'Smart CheckOut'. At the top, there is a status bar with the time '12:01' and various icons. Below the status bar is a red header with the text 'Smart CheckOut'. The main content area has a light pink background. In the center, there is a white square containing a black silhouette of a person's head and shoulders, with the word 'ADMIN' written in black capital letters on a white rectangular background. Below this icon, the text 'Admin Login' is displayed in a bold, black font. Underneath, there are two input fields: the first contains the text 'admin' and the second contains five dots, indicating a password field. Below these fields is a large, red rectangular button with the word 'LOGIN' in white capital letters. At the bottom of the screen, there is a black bar with three white horizontal lines, representing the Android navigation bar.

12:01

Smart CheckOut

ADMIN

Admin Login

admin

.....

LOGIN

Snapshots 5.5:Admin Login

CONCLUSION

The proposed project is feasible as it is built on technology that already exists. We have improved it by eliminating the long waiting time of queue at the time of billing, secured it by locking after completion of shopping and made the app available to all customers. Smart and hassle free checkout in supermarket is reliable and easy to use app. This app helps customer to save time in his busy schedule. This is the app which can be used by the customers and admin and floor manager which is user friendly for everyone. The development of this project helps everyone.

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- [6] <https://developer.android.com/guide/topics/ui/menus>[4]
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